

Assignment: Hybrid RAG – Intelligent Question Answering with PDF using EnsembleRetriever and Re-ranking

Course: Generative AI / Large Language Models (4th Year B.Tech)

Topic: Hybrid Retrieval-Augmented Generation (RAG)

Difficulty: Advanced

Objective

In this assignment, students will design and implement an Enterprise-grade Hybrid Retrieval-Augmented Generation (Hybrid RAG) system capable of answering user questions from one or more PDF documents.

The system must combine semantic search and keyword search using LangChain's EnsembleRetriever, followed by Cross-Encoder Re-ranking to improve retrieval quality before passing the context to a Large Language Model (LLM).

Students will implement the complete RAG pipeline including:

- PDF Loading
- Document Chunking
- Embedding Generation
- Vector Database Creation
- Dense Retrieval
- Sparse Retrieval (BM25)
- Hybrid Retrieval using EnsembleRetriever
- Cross-Encoder Re-ranking
- Prompt Engineering
- Question Answering
- Performance Evaluation

Dataset

Students may choose **one or more PDF documents** from any domain.

Suggested domains:

- Research Papers
- Company Annual Reports

- Product Manuals
- Technical Documentation
- Government Policies
- Medical Reports
- Books
- University Notes

Business Problem

Organizations often store knowledge in hundreds or thousands of PDF documents. Traditional keyword search fails to understand semantic meaning, while embedding-based retrieval alone may miss exact keyword matches.

Your task is to build an **Enterprise Knowledge Assistant** that combines:

- Dense Retrieval
- Sparse Retrieval
- EnsembleRetriever
- Cross-Encoder Re-ranking
- Large Language Model

to provide highly accurate and context-aware answers from PDF documents.

Mandatory Technologies

Students must use:

- Python
- LangChain
- Hugging Face Embeddings
- FAISS or ChromaDB or Any VDB
- BM25Retriever
- EnsembleRetriever
- Cross Encoder Re-ranker
- OpenAI / Gemini / Hugging Face LLM
- Streamlit and FastAPI Deployment

Assignment Tasks

Part 1: PDF Processing

Part 2: Embedding Generation & Vector Database

Part 3: Sparse Retrieval using BM25

Part 4: Dense Retrieval

Part 5: Hybrid Retrieval using EnsembleRetriever

Part 6: Question Answering Pipeline

Part 7: Prompt Engineering

Part 8: Performance Evaluation

Evaluate the Hybrid RAG system.

Compare the following systems:

System	Retrieval Method
Model 1	BM25 Only
Model 2	Dense Retrieval Only
Model 3	EnsembleRetriever
Model 4	EnsembleRetriever + Re-ranking

Evaluate using:

- Context Precision
- Context Recall
- Answer Relevancy
- Faithfulness
- Latency
- Response Time

Present the comparison in tables and charts.

Part 10: User Interface (Bonus)

Develop a Streamlit application.

Features:

- Upload pdf
- Upload API Keys
- Select Model

- Ask questions
- Display retrieved chunks
- Show re-ranked chunks
- Display final answer
- Display source documents
- Show retrieval pipeline

Business Insights

Assume you are an AI Engineer building an Enterprise Knowledge Assistant.

Answer the following:

1. Why is Hybrid Retrieval more effective than Dense Retrieval alone?
2. What role does EnsembleRetriever play in improving retrieval quality?
3. Why is BM25 still important in modern RAG systems?
4. Why is Cross-Encoder Re-ranking necessary after retrieval?
5. Which component contributed the most to answer quality?
6. What challenges did you face while implementing Hybrid RAG?
7. How can this architecture be used in healthcare, finance, legal, and education domains?